

Study Questions: Learning from Failure and Success

Analytical Questions

1. Distinguish between parameter learning and structure learning in Active Inference. Give an invention example of each and explain why structure learning is both more valuable and more difficult.
2. Describe the five-step post-mortem framework. Why is "establishing the facts" the necessary first step before "attributing causes"?
3. What is premature attribution, and why is it the most common failure in post-mortem analysis? How does it relate to the cognitive biases discussed in Module 04?
4. Explain the purpose of a failure mode catalog. How does it differ from simply remembering past failures?
5. Why is learning from success often neglected? What specific information does success analysis provide that failure analysis does not?
6. Define transfer learning in the context of invention. What is the role of abstraction in making lessons transferable?
7. Describe the five levels of the abstraction ladder (from specific observation to meta-principle). At which level does a lesson become maximally transferable? At which level does it risk becoming too vague to be actionable?
8. How does Toyota's andon cord system implement Active Inference learning principles at an organizational scale? What makes it more effective than periodic quality reviews?
9. Explain the difference between learning "what happened" and learning "why it happened." Which is more valuable for preventing future failures?
10. How can an inventor determine whether a successful outcome was due to correct model parameters (genuine understanding) or to luck and uncontrolled variables?

Applied Questions

1. You built a waterproof speaker prototype. During testing, the speaker survived submersion but produced distorted sound at high volumes. Conduct a post-mortem: establish facts, compare to predictions, attribute causes, and extract lessons. Classify each lesson as parameter or structure learning.
2. An inventor has completed 20 prototype iterations of a children's toy. She has records of each iteration but has never conducted a formal post-mortem. Design a retrospective analysis process she could use to extract lessons from all 20 iterations at once.
3. Your failure catalog shows that three of your last five prototypes have failed due to adhesive joint failure. Use the abstraction ladder to turn this specific pattern into a transferable design principle.
4. A startup's product works perfectly in testing but fails frequently in customer hands. What does this reveal about the relationship between test environment learning and real-world learning? How should the startup update its generative model?
5. Compare the learning value of a dramatic, catastrophic failure vs. a subtle, intermittent failure. Which is easier to learn from? Which is potentially more dangerous?
6. You invented a successful product — a collapsible shopping bag that is selling well. Conduct a success analysis: what factors contributed to success? Which were in your model? Which might be luck?
7. The WD-40 story describes 39 "failures" leading to a successful 40th formulation. Design a learning protocol that Norm Larsen could have used to ensure maximum information extraction from each failed formulation.
8. An aerospace company's failure mode catalog contains 5,000 entries from 40 years of testing. A new engineer is designing a wing component. How should they use the catalog? What challenges arise from a catalog this large?

9. You discover that your prototype works but you cannot explain why — the mechanism through which it achieves its function is unclear. Is this a learning success or a learning failure? What are the risks of deploying an invention you do not fully understand?
10. Design a one-page "learning template" that you would fill out after every prototype test. Include sections for facts, predictions, prediction errors, cause attribution, parameter updates, structure updates, and transferable lessons.